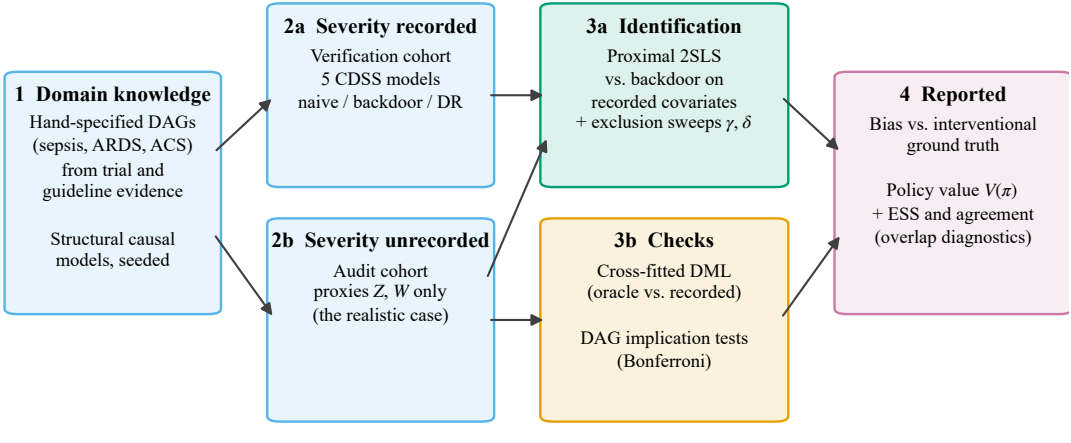

Ground truth at every stage: $\mathbb{E}[Y \mid \mathrm{do}(T = t)]$ computed by intervening on the same mechanisms
1 Domain knowledge
Hand-specified DAGs (sepsis, ARDS, ACS) from trial and guideline evidence
Structural causal models, seeded
2a Severity recorded
Verification cohort
5 CDSS models
naive / backdoor / DR
2b Severity unrecorded
Audit cohort
proxies Z, W only
(the realistic case)
3a Identification
Proximal 2SLS
vs. backdoor on
recorded covariates
+ exclusion sweeps γ, δ
3b Checks
Cross-fitted DML
(oracle vs. recorded)
DAG implication tests
(Bonferroni)
4 Reported
Bias vs. interventional ground truth
Policy value $V(\pi)$
+ ESS and agreement
(overlap diagnostics)